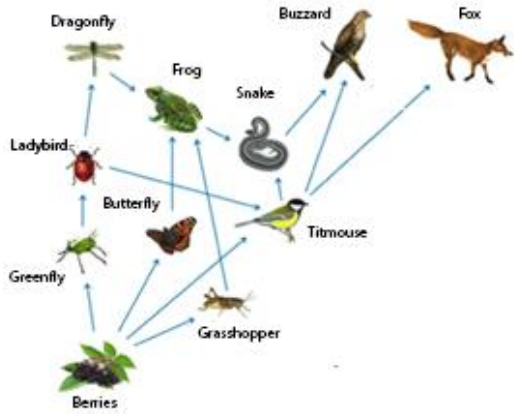


Question				Marking details		Marks available					
						AO1	AO2	AO3	Total	Maths	Prac
8./1.	(a)	(i)		Kingdom:	Animalia	1	2	3			
				Phylum:	Arthropoda						(1)
				Class:	Insecta						
				Order:	Hemiptera						
				Family:	Aphididae						
				Genus	<i>Acyrtosiphon</i>						(1)
				Species	<i>pisum</i>						(1)
				Binomial name	<i>Acyrtosiphon pisum.</i>						
Ignore case											
Allow <i>Acyrtosiphon pisum</i> reversed for (1)											
		(ii)	I	No confusion / unique name / universal / same in every language		1			1		
			II	Same genus/Vanessa/same first name		1			1		
	(b)	(i)	I	Migration		1			1		
			II	Temperature changes/weather/food availability Accept: breeding/better resources Do not accept: better conditions		1			1		
		(ii)		Larger population (than the mean last year) (1) Peaked earlier (1) Accept: population increased (1) Do not accept: population increased and decreased				2	2	2	
		(iii)		Similar shape line to the right of the dotted line (1) Smaller amplitude (1)			2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
HT		(iv)		As Red Admiral population increases so does the frog population (1) EITHER The population line for frogs lags behind that for the RA (1) because there is a delay before the frog population responds to the RA population change (1) OR The number of frogs never increases to the same value as the RA (1) because prey always outnumber the predator (1)		3		3		
FT	(c)	(i)		greenfly → <u>ladybird</u> (1) → <u>dragonfly</u> (1) → frog berries → <u>grasshopper</u> (1) 2 names in box (0)		3		3		
HT		(i)		One food chain drawn from berries (1) Second food chain drawn from berries (1) Third linked food chain (1) 						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
FT		(ii)		Rabbits and/or mice will decrease (1) Less food for foxes (1) fox numbers decrease (1) OR Rabbits and/or mice will decrease (1) Foxes (compete with buzzards to) feed on titmice (1) fox numbers decrease (1) OR Less food for rabbits and/or mice (1) Numbers decrease/die out (1) Less food for foxes/numbers decrease (1)			3	3		
FT HT		(iii) (ii)		Frogs occupy more than one food chain (1) which consist of different (numbers of) living things (1) OR Frogs have more than one food source / eat many things (1) In different (length) food chains (1) OR The frog is found in multiple food chains (1) Changes level depending on the chain (1) OR Eats primary and tertiary consumers (2) Treat secondary as neutral Do not accept they are omnivores			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
FT HT	(d)			shape (only one organism per level) (1) labelling of 4 levels – must include berries, buzzard and two other correct (1) order must start with berries and end with buzzard (1) buzzard snake frog grasshopper berries buzzard snake frog butterfly berries buzzard titmouse ladybird greenfly berries If 4/6/7 levels shown (2) only If 3 levels shown (1) only If sequence of arrows (2) max Sequence of lines / sequence of arrows, but one in wrong direction (1) only If: buzzard snake / titmouse frog/ladybird grasshopper/ butterfly /greenfly berries (2)			3	3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				berries grasshopper / butterfly / greenfly frog/ladybird snake/titmouse buzzard (2)						
FT HT	(e)			383 (1) 3368 (1) =11.37 / 11.4 / 11 (1) 0.11 / 0.1137 (2)	1	2		3	3	
				Question 8 & 1 total	6	9	10	25	5	